

Scholars High School Physics 1: Mechanics (4926)

Andrew Sayler

Fri

Sep 11, 2026 - Mar 12, 2027

7:30 PM - 9:00 PM ET (4:30 PM - 6:00 PM PT)

Homework

Week: **1** 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

Homework: Week 1



Readings

You have completed **10** of **10** challenge problems.

Your writing problem response has been graded.

Past Due **Sep 18**.

[Request Writing Problem Extension](#)

Week 1 Transcript: [Fri, Sep 11](#)

Challenge Problems

Problem 1 (43541)



In addition to this homework set, don't forget that there is a prereading to do before the next class (and there will be one before all the remaining classes in this course). It's important to do the prereading to be prepared for class. You can find it on the class homepage.

Problem:[Report Error](#)

You're not going to understand everything right away. Please ask your questions on our class message board! When you ask questions, other students benefit when they read and participate in the conversation. Instructors can also teach better when you ask questions, since it shows your thought process.

Below is the AoPS advice on how to ask a good question:



AoPS School Handbook wrote:

How To Ask A Good Question

Most often, we ask others questions to get answers. But at AoPS, we won't just tell you what the next step is or what you need to do in order to get a perfect score. Instead, we'll help you understand the material and ask questions that help you develop your own critical thinking skills. This will allow you to solve the problem on your own.

Some students may find this frustrating because the homework is due that day, and they just want to be told what to do. Please understand that you will not learn as much this way, and we will not just give you the answer or solution.

Moderate Level Question-Asking: Be Specific

Statements like "Please help" or "I'm stuck" are very hard to answer. After all, there aren't any details there! These statements will generally get a response of "What do you need help with?" or "What are you stuck on?" or "What don't you understand?"

In order to get help, think carefully about what you're are stuck on and try to tell us about it in your question. Some examples are: "I don't understand what the question means when it says ____."

"Can someone explain how (some property) applies to this problem?"

Expert Level Question-Asking: Providing Background

In addition to asking specific questions, great questions include some background. If we know about what you have already tried or how you're thinking about the problem, it will help us give better, more relevant tips. Expert-level questions look like this:

"I tried (some approach), but now I'm stuck. Can someone give me a hint?"

"It seems like we can use (some property) on this problem. I can tell how it applies __ *but I'm stuck on how it works with* __. Any tips?"

Knowing what you are having trouble with is an important skill to develop. To help us assist you in the best way, try to figure out specifically what is stopping you from moving forward. Try to make your question as specific as possible. We are here to help you, but we need some information to directly respond to your inquiry.

Getting Started

Regarding homework questions, AoPS has a general policy that students should get the discussion started. If you tell us you have no idea how to start, we will prompt you to come up with an idea to get the ball rolling. It doesn't even have to be a good idea, but looking at a problem and coming up with ideas is an important problem-solving skill that we want you to practice. After you get in the habit of coming up with ideas, then you can sort out the good ones.

[source](#)

Select the **three** statements that agree with the handbook's viewpoint on asking questions, as applied to a physics class.

a) Whoever answers the question should tell you the answer to the problem you're trying to solve.

b) Asking questions can be part of the process of solving a problem.

c) "How do I get started on this problem?" is an expert level question

d) Questions should communicate what specific part of a problem you are stuck on.

e) It's often helpful to start your question about a problem by saying what laws of physics you think are relevant.

f) If you're stuck, it's the instructor's responsibility to tell you the first step on a problem to get you going.

Solution:

a) Whoever answers the question should tell you the answer to the problem you're trying to solve.

b) Asking questions can be part of the process of solving a problem.

c) "How do I get started on this problem?" is an expert level question

d) Questions should communicate what specific part of a problem you are stuck on.

e) It's often helpful to start your question about a problem by saying what laws of physics you think are relevant.

f) If you're stuck, it's the instructor's responsibility to tell you the first step on a problem to get you going.

(a) The person answering the question should not tell you the answer. If they did that, you'd lose all the value of the problem, which comes from working out the answer yourself. Instead, the answerer should give some advice and help you get unstuck so you can answer the problem yourself.

(b) The handbook says "Instead, we'll help you understand the material and ask questions that help you develop your own critical thinking skills. This will allow you to solve the problem on your own." So yes, asking questions is part of the process of solving a problem on your own. Although you get help when you ask a question, you're still responsible for coming up with the solution.

(c) "How do I get started?" is not a specific question, so it's not an expert level question. Instead, you can say what you've already done to get started on a problem or what principles you think apply to a problem when you aren't sure how to approach a problem and want to ask about it.

(d) The handbook identifies "Be Specific" as the most important part of moderate level question-asking, so yes, questions should be specific.

(e) The handbook doesn't say anything about the laws of physics, but we can infer from what the handbook does say that stating what laws of physics you think are relevant is a good idea. The handbook talks about saying what mathematical approaches or properties you think are important to a problem. Thinking about what laws of physics to apply is an equivalent first step in a physics class.

(f) If you're stuck, the handbook says that instructors will give you prompts that may help you, but it's ultimately your responsibility to get going on the problem. You might not be able to solve the problem right away, but you can at least do something to get started, like state the relevant physics concepts, review what information you're looking for, identify some information you would like to find to help you (even if you don't know how to find it), or restate the problem in your own words.

Your Response(s):

Ⓔ b) Asking questions can be part of the process of solving a problem., d) Questions should communicate what specific part of a problem you are stuck on., e) It's often helpful to start your question about a problem by saying what laws of physics you think are relevant.

**Problem:**[Report Error](#)

One of the core skills physicists learn is to decide how much complexity to put in their models. They decide this based on the type of problem they're working on and the accuracy they need.

In this problem, we'll practice thinking about when we can productively use the most basic models and when we need more complicated models.

Almost nothing has a perfectly constant velocity, but some things are close.

Of the situations below, which is the best one to be modeled as motion with constant velocity?

note on (d)

note on (e)

a) Observing the flight of a hummingbird for three minutes as it collects nectar from around a meadow.

b) Tracking the flight of an airplane on a long flight for three uneventful minutes in the middle of the flight.

c) Watching high-speed video of a violin string oscillating back and forth as the violin is played.

d) Modeling the motion of birds flying in a murmuration.

e) Finding the total distance traveled by a soccer ball over an entire game.

Solution:

The best choice is

b) Tracking the flight of an airplane on a long flight for three uneventful minutes in the middle of the flight.

(a) (hummingbird)

A hummingbird will frequently stop, then fly again, and repeat its stop-go cycle many times in three minutes. It will fly at different speeds in those three minutes depending on where it's going, so you will lose a lot of detail by assuming constant speed.

(b) (airplane)

In the middle of a long flight, an airplane flies at approximately constant speed — the fastest speed it safely and efficiently can. There may be some small speed variations, but they will be inconsequential for most purposes.

(c) (violin string)

Since the string is going back and forth, it repeatedly speeds up and slows down. Its speed is not at all constant.

(d) (birds)

The birds repeatedly change direction throughout the short GIF, indicating their velocity is not close to constant.

(e) (soccer)

Over a soccer game, the ball stops and starts and changes direction many times, so a constant velocity model would not be appropriate.

Hint(s): For each option, ask whether the speed and direction stay roughly the same over the whole time interval, or whether the motion visibly speeds up, slows down, stops, or changes direction.

Your Response(s):

☒ b) Tracking the flight of an airplane on a long flight for three uneventful minutes in the middle of the flight.

Problem 4 – Correct! – Score: 5 / 7 (43554)

**Problem:**[Report Error](#)

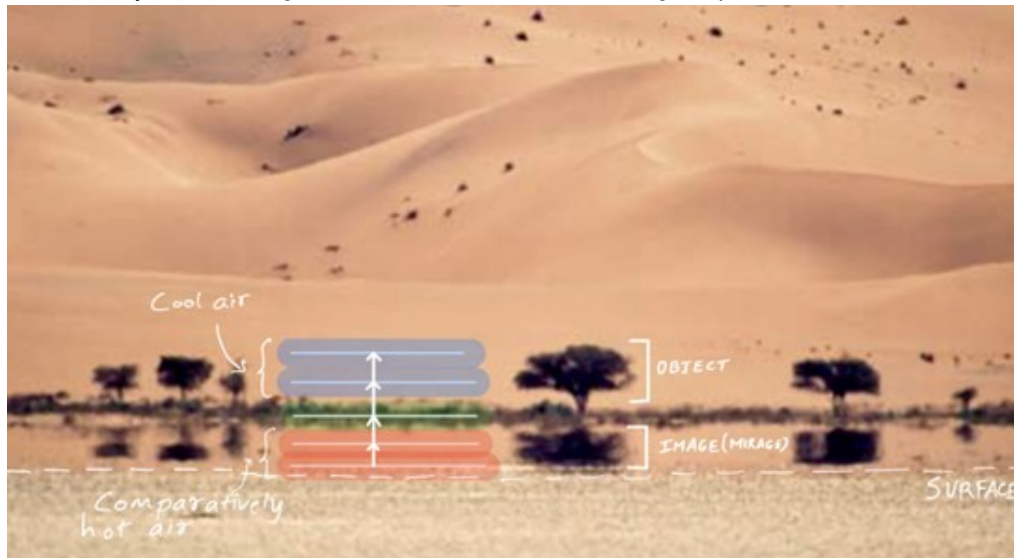
On hot days you can sometimes see *mirages*.



PhysicsStuff.com

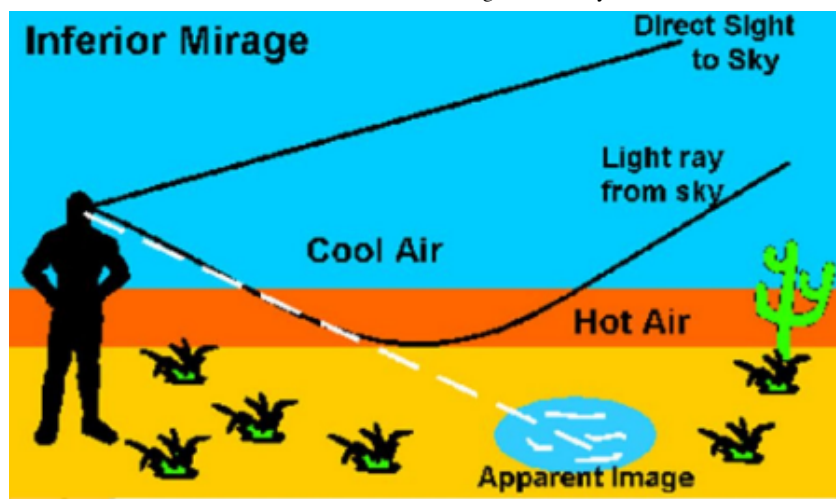
The reflection of people walking can make it look like there's a puddle of water on the ground, but there isn't.

A mirage happens when the air just above the ground is much warmer than the air higher up.



PhysicsStuff.com

Under these conditions, light will bend down and then up again, creating the mirage.



Madhav University

note on diagram

Remember from class that light takes the path of least time between its starting and end points.

What does this imply about the speed of light through air?

Choose one.

- a) The speed of light is always equal to c by definition, so it doesn't imply anything.
- b) The speed of light is faster in hot air than in cold air.
- c) The speed of light is faster in cold air than in hot air.
- d) The speed of light is slower in air than in a vacuum, but the same regardless of the air temperature.
- e) Nothing; the light is pulled down by gravity, not the temperature of the air.

Solution:

The answer is **b) The speed of light is faster in hot air than in cold air**.

The shortest path from an object to you is a straight line. If light takes a longer path, such as a path that bends, it must be taking that path because, although it's longer, the average speed of light is faster along that path.

Since light dips down into the hot air to get to you quickly, we conclude that the speed of light is **faster in hot air** than in cold air.

This is because hot air is less dense than cold air — there are fewer molecules of air per cubic meter in hot air. With fewer molecules, there are fewer interactions to slow light down.

Hint(s): This is on the right track, but if light is faster in cold air, a straight path, not a bent one, would be fastest. Light could travel directly to you through the cooler air without bending down. The path would be shorter and have a higher speed than one that bends down low.

The light bends down to travel through the hot air for a while, going some extra distance. This still doesn't take extra time, though, so is light going faster or slower through hot air?

The shortest path to you is a straight line, but the light takes a curved path, which is longer. If a longer path takes less total time, then the light must be travelling faster on average on the curved path than it would on a straight path.

Your Response(s):

- ☹ c) The speed of light is faster in cold air than in hot air.
- 😊 b) The speed of light is faster in hot air than in cold air.

Problem:[Report Error](#)

We will use *normalized scientific notation* for both large and small numbers. In this notation, any number can be represented by a number (called the "coefficient") greater than 1.0 and less than 10, multiplied by ten to some power. For example, here are some numbers written in both the ordinary decimal notation and in normalized scientific notation.

note on the word "normalized"

decimal notation	normalized scientific notation
350	3.50×10^2
10,005	1.0005×10^4
14.3	1.43×10^1
578,240.92	5.7824092×10^5
0.00037	3.7×10^{-4}

If you would like a tutorial on scientific notation, there is one available on the class homepage.

We also want to report the results of our calculations to an appropriate number of *significant figures*.

For example, if a problem says that something travels 2.2 m in 0.91 s, it's reasonable to give the object's speed as 2.4 m/s, but the precise calculation giving $\frac{2.2 \text{ m}}{0.91 \text{ s}} = 2.417582 \dots \text{ m/s}$ has digits that aren't meaningful, since the original measurements are not known to that level of detail.

In a lab report or scientific document, adding the extra digits could confuse the reader by making them think your data was far more precise than it really was. As a guideline, give your calculations to the same number of digits as used in the data.

The speed of sound is usually faster in solids than in air. It's especially fast in beryllium, a rare metal that is exceptionally stiff and light. Beryllium is sometimes used when weight is extremely important, for example in satellites. A sample of beryllium looks like this:



By Alchemist-hp = Alchemist-hp (pse-mendelejew.de) - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=4283181>

Suppose a student wants to find the speed of sound v through a sample of beryllium. Using a micrometer (a device for precise distance measurements) and electronic timing equipment, they measure that sound travels $1.08 \times 10^{-1} \text{ m}$ in $8.51 \times 10^{-6} \text{ s}$. What speed of sound do they compute?

Give your answer in normalized scientific notation and to an appropriate number of significant figures.

$$v = 1.27 \times 10^4 \text{ m/s}$$

Solution:

The speed of sound in beryllium is $v = 1.27 \times 10^4 \text{ m/s}$.

The speed is the distance traveled divided by the time needed, or

$$v = \frac{1.08 \times 10^{-1} \text{ m}}{8.51 \times 10^{-6} \text{ s}}.$$

We can divide the coefficients to get $\frac{1.08}{8.51} = 0.1269095 \dots$. However, because the problem gave us three significant figures for both the distance and time, we should only use three significant figures in our answer as well, so we round this to 0.127

Hidden Content

Doing the division in the exponents, we have

$$\begin{aligned} \frac{10^{-1}}{10^{-6}} &= 10^{-1-(-6)} \\ &= 10^5. \end{aligned}$$

That makes the speed of sound in beryllium

$$v = 0.127 \times 10^5 \text{ m/s}.$$

However, this is not normalized scientific notation because the coefficient 0.127 is not greater than or equal to one. To fix this, we need to take one power of ten from the exponent and move it to the coefficient.

$$\begin{aligned} v &= 0.127 \times 10^5 \text{ m/s} \\ &= (0.127 \cdot 10) \times \left(\frac{1}{10} \cdot 10^5 \right) \text{ m/s}. \end{aligned}$$

Simplifying both the coefficient and the exponent, we have

$$v = 1.27 \times 10^4 \text{ m/s}.$$

This is around 40 times faster than the speed of sound in air on a normal day. Other solids usually have speeds of sound that are faster than the speed of sound in air, which means that sound can travel to you faster through a brick wall than it can through the air next to the wall. However, beryllium has an unusually fast speed of sound, even for a solid, largely because its atoms are very light (it is the fourth-lightest element) and can shake back and forth very quickly to spread sound waves. We will learn more about the relationship between mass and oscillations later in the course.

Hint(s): Both the distance and time the student measured have three digits in them, so your answer should have three digits in the coefficient. You can divide the student's distance measurement by their time measurement to get the speed, but then make sure you put the answer in normalized scientific notation.

Your Response(s):

⊕ $v = 1.27 \times 10^4 \text{ m/s}$

**Problem:**[Report Error](#)

This question is to help make the message boards useful for the entire class by practicing writing titles for questions. The better the titles we use, the more everyone can gain by browsing the message boards.

On AoPS message boards, if you ask a question, you'll make a title for that question first. A good title should ask a specific question so that people reading the thread will know what the question is about before they click through to the question itself. That means it should be a short summary of the main idea behind your question.

You just solved a problem on significant figures and scientific notation. Suppose a student was working on this problem, but had never heard of significant figures before and didn't understand them based on the problem alone.

The student asks a question about this homework problem on the message boards. Rank the following titles for their question from best (top) to worst (bottom).

- 1 In wk1, p. 3, how do I know how many significant figures there are in a number?
- How do I start week 1, problem 3?
- worst: Please help me!

Solution:

The correct ordering is:

- best: In wk1, p. 3, how do I know how many significant figures there are in a number?
- How do I start week 1, problem 3?
- worst: Please help me!

"Please help me!" is not a good title for a thread. It doesn't convey any information about the question, and so neither classmates nor course assistants will know what the thread is about before clicking on it, aside from that it's someone looking for help. Titles such as "Why is this wrong?", "I'm stuck!", or "I got half way." are similarly not informative.

"How do I start week 1, problem 3?" is a better title, but could still be improved. It identifies the specific question, although most people will not remember off the top of their head what "week 1, problem 3" is, so it's not as informative as it could be. Also, "How do I get started?" is not a specific question about a problem. If you read a problem and don't know how to get started, you can still figure out *something* about the question. For example, you might be able to say things like:

- "It looks like the difference between speed and velocity is important in this question."
- "I think this question is asking for a conceptual or qualitative analysis, not a precise calculation with numbers or symbols, is that right?"
- "This problem looks somewhat similar to the problem about a rocket that we did in class, but it's different because the rocket is moving in two dimensions now, not one. Will I need to bring in some extra mathematical tools to solve this problem?"

These are the sorts of things you can figure out even if you are completely stuck on solving the problem and can't take the first step. It is

better to identify what thoughts you have on the problem and why you feel stuck than to simply say, "I'm stuck and can't start." It's much easier for course assistants and classmates to help you if you can say what you've already considered, even if you can't get started on any solution path.

"In wk 1, p. 3, how do I know how many significant figures there are in a number?" is the best title. It makes it clear that the student has figured out *why* they are stuck. They expect to understand what a concept like significant figures is before they solve problems on it. They realize that significant figures are something you can count in a number, but don't know the rules for how to count them. Other students could read this title and immediately realize they had the same difficulty, and course assistants can read it and know right away what sort of help the thread is asking for.

Your Response(s):

🗨 In wk1, p. 3, how do I know how many significant figures there are in a number?, How do I start week 1, problem 3?, Please help me!

Problem 7 – Correct! – Score: 7 / 7 (43552)



Problem:

[Report Error](#)

This problem is practice with the ideas of speed, distance, and time.

Suppose two cars begin driving towards each other from 150 mi apart.

The first car drives at 40 mi/h and the second car drives at 60 mi/h.

How long will it take the cars to meet?

1.5 h

Solution:

The distance between the cars is decreasing at the rate $60 \text{ mi/h} + 40 \text{ mi/h} = 100 \text{ mi/h}$.

The distance between the cars is 150 mi, so for this distance to decrease to zero will take a time

$$\frac{150 \text{ mi}}{100 \text{ mi/h}} = 1.5 \text{ h}.$$

Hint(s): Switch into the reference frame of the first car. How fast is the second car going in this reference frame? In this reference frame, the second car travels the entire 150 mi and the first car doesn't go anywhere. Another way to think of it is to find how fast the distance between the cars is shrinking. If you know how far apart the cars are to begin with and how fast the distance between them is decreasing, you can use division to find the time it will take for the cars to meet.

Your Response(s):

🗨 1.5 h

**Problem:**[Report Error](#)

This problem is more practice with the mathematical relation between speed, distance, and time.

A driver wants to complete a 60.0 mi trip at an average of 60.0 mi/h. They've already completed the first 30.0 mi at an average of 70.0 mi/h. What speed should the driver set for the second 30.0 mi to make the average come to 60.0 mph for the entire trip?

Give your answer as a decimal.

52.5 mi/h

Solution:

The average speed needed in the second half is 52.5 mi/h.

The first 30 mi took

$$\frac{30 \text{ mi}}{70 \text{ mi/h}} = \frac{3}{7} \text{ h.}$$

That means $\frac{4}{7}$ h remain for the second 30 mi, so in the second half, the driver needs an average speed of

$$\frac{30 \text{ mi}}{\frac{4}{7} \text{ h}} = 52.5 \text{ mi/h.}$$

The trap in this problem is to assume the answer is 50 mi/h because the average of 50 and 70 is 60. However, since the driver is going faster in the first half of the trip than the second half, they spend less time in the first half. If they drive the first 30 mi at 70 mi/h and the second at 50 mi/h, they will spend more time driving at 50 mi/h than at 70 mi/h, so the average comes out to be closer to 50 mi/h than to 70 mi/h.

One way around this trap is to think in terms of hours per mile. Since the driver is driving for the same number of miles in the faster and slower portions of the trip, we can average the number of hours per mile in the first half of the trip with the number of hours per mile in the second half of the trip to get the average hours per mile. Using the correct speeds we've already figured out, this comes to

$$\frac{\frac{1 \text{ h}}{70 \text{ mi}} + \frac{1 \text{ h}}{52.5 \text{ mi}}}{2} = \frac{1 \text{ h}}{60 \text{ mi}},$$

which confirms our answer.

Averaging 70 and 52.5 to get 60, as we've done here, is taking the *harmonic mean* of the two. The harmonic mean of a and b is defined to be $\frac{2}{\frac{1}{a} + \frac{1}{b}}$.

Hint(s): Find the total time the driver should take on the entire 60 mi trip. Then find how long it takes them to travel the first 30 mi at 70 mi/h. Then use subtraction to find the time left to complete the last 30 mi. Then compute the speed the driver needs to maintain for the last 30 mi.

Your Response(s):

52.5 mi/h

**Problem:**[Report Error](#)

In this problem, you'll practice *estimation* and *unit conversions*. These are both essential skills that we'll use throughout the course.

Estimate the average speed at which human hair grows in miles per hour.

There is no one answer to this question — different hairs grow at different rates, and grow unsteadily. Still, there is a range which is approximately correct. To be marked correct on this question you should be within one **order of magnitude** of our estimate.

In physics, when we estimate, we don't just guess our final answer. Instead, we start with something we know and work from there. Try to estimate the speed that hair grows based on some life experience you have, and use whatever units are most convenient. Then convert your units to miles per hour.

Here are a few conversion factors that could help:

$$2.54 \text{ cm} = 1 \text{ in}$$

$$12 \text{ in} = 1 \text{ ft}$$

$$5280 \text{ ft} = 1 \text{ mi}$$

You may also want to use conversion factors related to hours, days, months, or years.

Enter your answer using normalized scientific notation, meaning that the coefficient should be greater than or equal to one and less than ten.

$$\text{hair growth speed} \approx \boxed{8} \times 10^{\boxed{-8}} \text{ mph}$$

Solution:

The growth rate of hair is $\boxed{2 \times 10^{-8} \text{ mph}}$.

We should begin with some experience we have with hair growth. For example, how often do you get a haircut, and how much hair do you cut off each time?

Based on your experience, you'll probably estimate that hair grows around one inch per month.

Perhaps you haven't had a haircut recently and don't know how much hair grows in a month. You can probably still think of *something* you know that will let you figure it out. For example, your friend shaves their head. How long do you think it will be before they have hair down to their shoulders? If you don't know, can you guess between the options:

- 1) one day
- 2) one week
- 3) one month
- 4) one year
- 5) ten years

Yes, you can probably guess that it would take at least a year to grow hair down to your shoulders after shaving your head; ten years is too long, since children much younger than that can have hair that long, and one week is definitely too short. One month is too short, too, as you'll remember if you've ever had a friend who shaved their head. This shows that even if you can't get the answer exactly, you can get it approximately.

Now let's convert the figure of one inch per month into miles per hour. We can do this with a series of conversion factors.

$$\begin{aligned} \frac{1 \text{ in}}{1 \text{ month}} \cdot \left(\frac{1 \text{ ft}}{12 \text{ in}} \right) \cdot \left(\frac{1 \text{ mi}}{5280 \text{ ft}} \right) \cdot \left(\frac{1 \text{ month}}{30 \text{ days}} \right) \cdot \left(\frac{1 \text{ day}}{24 \text{ hr}} \right) &\approx 2 \times 10^{-8} \frac{\text{mi}}{\text{hr}} \\ &= 2 \times 10^{-8} \text{ mph.} \end{aligned}$$

In setting up this calculation, make sure that each conversion factor cancels out a previous unit that you don't want. For example, the first conversion factor we used is $\frac{1 \text{ in}}{1 \text{ month}} \cdot \left(\frac{1 \text{ ft}}{12 \text{ in}} \right)$. The reciprocal conversion factor is also a valid conversion factor. If we had used it, we would have $\frac{1 \text{ in}}{1 \text{ month}} \cdot \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)$. The reason we use the first version is that this way, there are inches (in) in the numerator of the first factor and in the denominator of the conversion factor, so they cancel, and we get rid of inches, which we didn't want in our final answer. With the second conversion factor, we'd get the awkward units of $\frac{\text{in}^2}{\text{month} \cdot \text{ft}}$. While not technically wrong units, these units don't get us any closer to finding miles per hour.

To allow for how your estimates will vary based on your own life experiences, your own hair, and your memories of haircuts and hair growth, we accept answers between $2 \times 10^{-7} \text{ mph}$ and $2 \times 10^{-9} \text{ mph}$ for this question.

Hint(s): Start from something you actually know, like how often you get a haircut and how much hair that removes, and set up conversion factors so each unwanted unit cancels against the next factor.

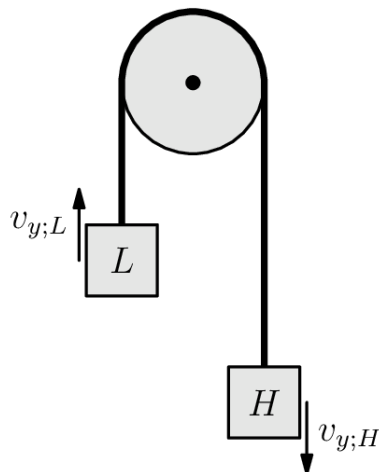
Your Response(s):

⊕ hair growth speed $\approx 8 \times 10^{-8} \text{ mph}$

Problem:[Report Error](#)

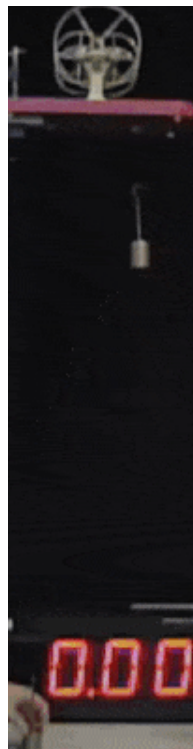
In this problem, you'll practice the notation and interpretation of speed and velocity.

Below is a sketch of an *Atwood machine*, an 18th century invention intended to measure the strength of gravity. The device consists of a pulley (a circular ring that can easily spin, but otherwise can't move around left/right or up/down), a rope that goes over the pulley, and weights on either side of the rope. The box H is heavier than the box L .



The weights are initially held with L down low and H up high, then released. H will start falling downward, and it will pull on the rope as it falls, which pulls L upward.

To help you visualize this motion, look at this GIF:



"Low Friction Atwood Machine," mittechtv, YouTube

(In this GIF one of the weights has some slight side-to-side wobble, which you should ignore. Model the motion as purely in the up/down direction.)

Let the velocity of the heavier box be $v_{y;H}$ and the lighter box $v_{y;L}$. The speeds are v_H and v_L . We define the upward direction to be

positive, so upward motion is a positive velocity.

One second after the boxes are released, we measure the velocities $v_{y;H}$ and $v_{y;L}$. (The boxes don't hit the ground or reach the pulley in this one second.)

Which statements below are true? Select two.

You do not need to find the values of $v_{y;H}$ or $v_{y;L}$, or make any quantitative measurements, to answer this question.

Note that statements (a) and (b) are about velocities. (c) through (e) are about speeds.

a)

$$v_{y;L} = -v_{y;H}$$

b)

$$v_{y;L} = v_{y;H}$$

c)

$$v_H > v_L$$

d)

$$v_H < v_L$$

e)

$$v_H = v_L$$

Solution:

a)

$$v_{y,L} = -v_{y,H}$$

b)

$$v_{y,L} = v_{y,H}$$

c)

$$v_H > v_L$$

d)

$$v_H < v_L$$

e)

$$v_H = v_L$$

(a)

The length of the rope can't change, so whenever one box falls by 1 cm, the other box rises by 1 cm. This means both boxes move the same distance in any given time, so they have the same speed. However, one moves up while the other moves down, so they move in opposite directions. That means they have opposite signs, so $v_{y,L} = -v_{y,H}$.

(b)

This is false because we've just seen that the velocities are opposites, not the same.

(c)

We've already found that the weights move with the same speed, so $v_H = v_L$. That makes (c) false.

(d)

This is false for the same reasoning as (c).

(e)

As we saw in (c), this is true, $v_H = v_L$. This equation says the speeds of the boxes are the same, even though they move in opposite directions.

Hint(s): The speeds of the weights are equal, but one moves up and the other moves down. Be careful about the difference between v , a speed, and v_y , a velocity. Speed is the magnitude (absolute value) of velocity.

Your Response(s):

☺ a)

, e)

$$v_{y,L} = -v_{y,H}$$

$$v_H = v_L$$

Problem:

Report Error

This problem will help you practice the relationship between velocity and distance traveled when the velocity changes and is represented on a graph.

Ayesha McGowan is a professional road racing cyclist.



Jeff Clark, <https://www.liv-cycling.com/global/campaigns/representation-matters-ayasha-mcgowan/23833>

In 2021, McGowan competed in a multi-day race near Avignon, France. The race took her into the western portion of the Alps mountain range. As you would expect, she rode relatively slowly when climbing steep mountains and very fast when coming down.

McGowan's bike has a computer that records its speed. After her rides, McGowan shares data on her speed versus time, along with some other data, publicly on the internet. Below is a plot of some of Ayesha's data during one day of the race. Time is indicated along the top. The total time was close to four hours.



How far did McGowan ride this day?

Your answer should be accurate to within 20 mi.

hint

McGowan rode miles

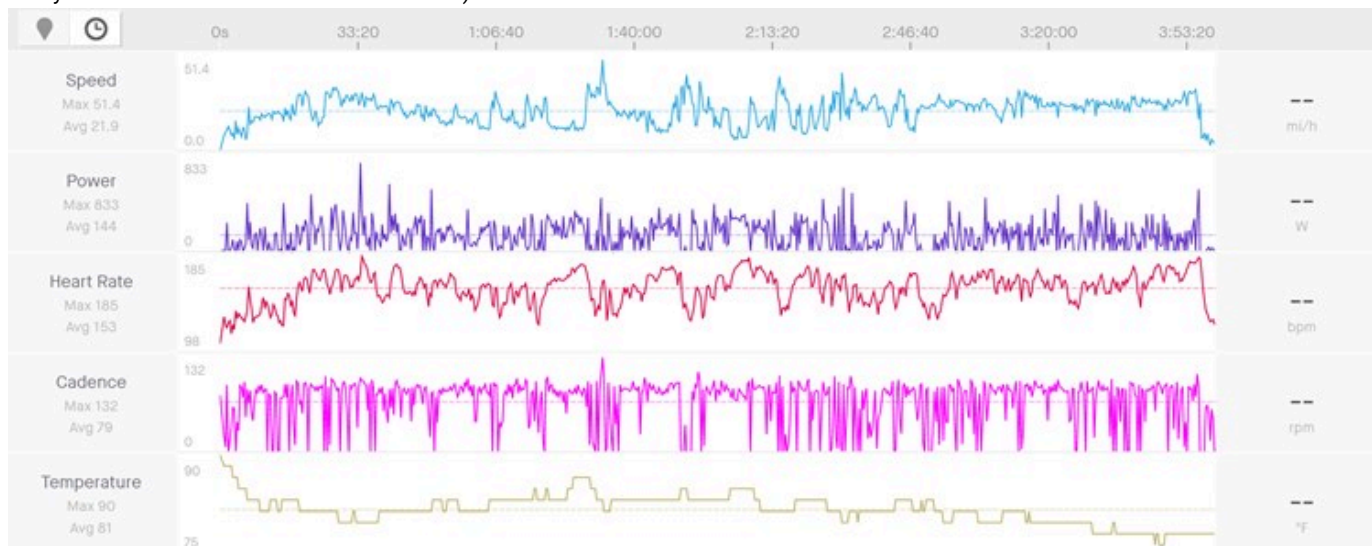
Solution:

McGowan's speed varied significantly throughout the ride. We could find her total distance traveled by finding the area underneath her speed versus time curve.

Another option is to estimate McGowan's average speed. She reached a maximum of $51.4 \frac{\text{mi}}{\text{h}}$ on a downhill, but most of the time, she rode around half that speed, and she rode somewhat slower when climbing hills.

We could guess that McGowan's average speed was in the range of $20 \frac{\text{mi}}{\text{h}}$ to $25 \frac{\text{mi}}{\text{h}}$. Multiplying that by the duration of the race – approximately four hours, we get an estimate of 80 mi to 100 mi. In the middle of that range would be 90 mi.

Here is the original plot, which includes McGowan's average speed of $21.9 \frac{\text{mi}}{\text{h}}$. (We digitally deleted the average for this question so that you would have the chance to estimate it.)



<https://www.strava.com/activities/5937921236/analysis>

McGowan's speed is sometimes above and sometimes below the average. The area between the average and her speed graph when she went faster than $21.9 \frac{\text{mi}}{\text{h}}$ is equal to the area between the average and her speed graph from times she went slower than 21.9 mi/h .

From the average speed of $21.9 \frac{\text{mi}}{\text{h}}$, we can calculate McGowan biked $21.9 \frac{\text{mi}}{\text{h}} \cdot 4.0 \text{ h} \approx 82 \text{ mi}$.

McGowan's computer calculated the total distance of the ride to be **87.63 miles**. We accept answers from **65 mi** to **120 mi** for this problem.

Hint(s): You only need to use the speed graph, along with the times printed along the top, to answer this question. The other data isn't necessary. Although the speed varies up and down, you can guess an average speed based on the graph. Her average speed seems to be in the neighborhood of **20 mi/h**.

Your Response(s):

☺ McGowan rode 90 miles

Problem 12 – Graded (43614)

**Problem:**[Report Error](#)

Take a look at the following dot diagram. The dots were made 1 s apart, and the scale is 1 : 1, so the dots' distance apart on your computer monitor is their distance apart in real life.

measurement note



a)

Did the object modeled by this dot diagram have positive or negative velocity? How do you know? If it's impossible to tell, explain why.

For the rest of the question, assume the object moved from left to right and that it never had the same speed at any two different times.

b)

What happened to the speed of the object over time? How do you know? Your answer can be qualitative here.

c)

What was the approximate average speed of the object over the entire time modeled by the dot diagram? Show your work in calculating it.

d)

Estimate the instantaneous speed of the object at the time of the second-to-last dot and explain how you did this.

Your Response:

a) It is impossible to know whether the dots have positive or negative velocity, as we don't know what direction the object is moving.

b) The speed increases over time. Since the time to get from one dot to another stays the same, but the distance travelled increases, the speed increases, as distance = speed/time.

c) We can see the positions of the dots, from left to right, are approximately 0, 1, 4, 9, 16, 25, 36. Therefore, the average speed is about $\frac{36}{6} = 6$ meters per second, given that position n and position $n + 1$ are one meter apart.

d) The instantaneous speed occurs between the 5th dot and the 7th dot. Those positions are 16 and 36, and it takes two seconds to go from one point to the other. Thus, the instantaneous speed is $\frac{36 - 16}{7 - 5} = 10$ meters per second, assuming that the distance between position n and position $n + 1$ is 1 meter.

Technical Score: 5.5 / 7**Style Score:** 1 / 1**Comments:**

(a)

You correctly explain that the direction of motion is unknown, so the sign of the velocity cannot be determined.

(b)

Your conclusion and your comparison of distances traveled during equal time intervals are correct. The formula at the end needs correction: distance = average speed $\times$ time. Your calculations in parts (c) and (d) correctly use distance divided by time to calculate average speed.

(c)

The numbers 0, 1, 4, 9, 16, 25, 36 describe the drawing's coordinates, but they do not establish distances in meters. The last gap spans $36 - 25 = 11$ coordinate units, while its physical length is about 3.9 cm. Use this information to determine the physical length of one coordinate unit, then convert the total distance before dividing by 6 s.

(d)

Choosing the fifth through seventh dots and using an elapsed time of 2 s is appropriate. The average speed over this interval estimates the instantaneous speed at the sixth dot. Your result carries forward the same distance-scale error from part (c), so apply the corrected conversion here as well.

Keep up the effort!

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